4425 Manchester Road
Kalamazoo, MI 49001
Phone 269 381-9666
Fax 269 381-9698
www.karlabs.com

City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : ***Ms. Amy Vail***

KAR Project No. :	***114264***
Date Reported :	***10/18/11***
Date Activated :	***10/04/11***
Date Due :	***10/18/11***
Date Validated :	***10/18/11***

Project Description : ***Analysis of two aqueous samples for NPDES permit.***

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 114264. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

David R. Alkema
David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *114264*

Date Reported : *10/18/11*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: *Analysis of two aqueous samples for NPDES permit.*

Sample ID : *"Plant Effluent, Composite"*

Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *10/03/11*
Sample Time : *0745*

Date Received : *10/04/11*
Sample Type : *aqueous*
KAR Sample No. : *114264-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, metals	Completed		EPA 30xx,200.x	10/05/11	DBL	
Antimony, total, low level	<1	ug/L	EPA 200.8	10/11/11	PML	
Arsenic, total, low level	1	ug/L	EPA 200.8	10/11/11	PML	
Barium, total, low level	20	ug/L	EPA 200.8	10/11/11	PML	
Beryllium, total, low level	<1	ug/L	EPA 200.8	10/11/11	PML	
Boron, total, low level	216	ug/L	EPA 200.7	10/14/11	DBL	
Cadmium, total, low level	<0.2	ug/L	EPA 200.8	10/11/11	PML	
Chromium, total, low level	<5	ug/L	EPA 200.8	10/11/11	PML	
Copper, total, low level	8	ug/L	EPA 200.8	10/11/11	PML	
Lead, total, low level	<1	ug/L	EPA 200.8	10/11/11	PML	
Nickel, total, low level	<5	ug/L	EPA 200.8	10/11/11	PML	
Silver, total, low level	<0.5	ug/L	EPA 200.8	10/11/11	PML	
Thallium, total, low level	<1	ug/L	EPA 200.8	10/11/11	PML	
Hardness	217	mg/L (as CaCO3)	EPA 130.2	10/10/11	PML	
Prior. Poll. BN by 625	See below		EPA 625	10/11/11	KTL	
Prior. Poll. acids by 625	See below		EPA 625	10/11/11	KTL	
Prep, SV Acid/BN	Completed		EPA 625	10/07/11	KTL	
1,2,4-Trichlorobenzene 625	<5	ug/L	EPA 625	10/11/11	KTL	
1,2-Diphenylhydrazine	<5	ug/L	EPA 625	10/11/11	KTL	
2,4,6-Trichlorophenol	<5	ug/L	EPA 625	10/11/11	KTL	
2,4-Dichlorophenol	<5	ug/L	EPA 625	10/11/11	KTL	
2,4-Dimethylphenol	<5	ug/L	EPA 625	10/11/11	KTL	
2,4-Dinitrophenol	<20	ug/L	EPA 625	10/11/11	KTL	
2,4-Dinitrotoluene	<5	ug/L	EPA 625	10/11/11	KTL	
2,6-Dinitrotoluene	<5	ug/L	EPA 625	10/11/11	KTL	
2-Chloronaphthalene	<5	ug/L	EPA 625	10/11/11	KTL	
2-Chlorophenol	<5	ug/L	EPA 625	10/11/11	KTL	
2-Methyl-4,6-dinitrophenol	<20	ug/L	EPA 625	10/11/11	KTL	
2-Nitrophenol	<5	ug/L	EPA 625	10/11/11	KTL	
3,3'-Dichlorobenzidine	<20	ug/L	EPA 625	10/11/11	KTL	
4-Bromophenyl phenyl ether	<5	ug/L	EPA 625	10/11/11	KTL	
4-Chloro-3-methylphenol	<5	ug/L	EPA 625	10/11/11	KTL	
4-Chlorophenyl phenyl ether	<5	ug/L	EPA 625	10/11/11	KTL	
4-Nitrophenol	<20	ug/L	EPA 625	10/11/11	KTL	
Acenaphthene	<5	ug/L	EPA 625	10/11/11	KTL	
Acenaphthylene	<5	ug/L	EPA 625	10/11/11	KTL	
Anthracene	<5	ug/L	EPA 625	10/11/11	KTL	
Benzidine	<50	ug/L	EPA 625	10/11/11	KTL	
Benzo(a)anthracene	<5	ug/L	EPA 625	10/11/11	KTL	
Benzo(a)pyrene	<5	ug/L	EPA 625	10/11/11	KTL	
Benzo(b)fluoranthene	<5	ug/L	EPA 625	10/11/11	KTL	
Benzo(ghi)perylene	<5	ug/L	EPA 625	10/11/11	KTL	
Benzo(k)fluoranthene	<5	ug/L	EPA 625	10/11/11	KTL	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *114264*

Date Reported : *10/18/11*

Attest: David R. Alkema

David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Composite"***

Sampled By : *AV of City of Cadillac, Water Resources*

Sample Date : *10/03/11*

Sample Time : *0745*

Date Received : *10/04/11*

Sample Type : *aqueous*

KAR Sample No. : *114264-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Bis(2-chloroethoxy)methane	<5	ug/L	EPA 625	10/11/11	KTL	
Bis(2-chloroethyl)ether	<5	ug/L	EPA 625	10/11/11	KTL	
Bis(2-chloroisopropyl)ether	<5	ug/L	EPA 625	10/11/11	KTL	
Bis(2-ethylhexyl)phthalate	<5	ug/L	EPA 625	10/11/11	KTL	
Butylbenzyl phthalate	<5	ug/L	EPA 625	10/11/11	KTL	
Chrysene	<5	ug/L	EPA 625	10/11/11	KTL	
Di-n-butyl phthalate	<5	ug/L	EPA 625	10/11/11	KTL	
Di-n-octyl phthalate	<5	ug/L	EPA 625	10/11/11	KTL	
Dibenzo(ah)anthracene	<5	ug/L	EPA 625	10/11/11	KTL	
Diethyl phthalate	<5	ug/L	EPA 625	10/11/11	KTL	
Dimethyl phthalate	<5	ug/L	EPA 625	10/11/11	KTL	
Fluoranthene	<5	ug/L	EPA 625	10/11/11	KTL	
Fluorene	<5	ug/L	EPA 625	10/11/11	KTL	
Hexachlorobenzene	<5	ug/L	EPA 625	10/11/11	KTL	
Hexachlorobutadiene	<5	ug/L	EPA 625	10/11/11	KTL	
Hexachlorocyclopentadiene	<5	ug/L	EPA 625	10/11/11	KTL	
Hexachloroethane	<5	ug/L	EPA 625	10/11/11	KTL	
Indeno(123cd)pyrene	<5	ug/L	EPA 625	10/11/11	KTL	
Isophorone	<5	ug/L	EPA 625	10/11/11	KTL	
N-Nitrosodi-n-propylamine	<5	ug/L	EPA 625	10/11/11	KTL	
N-Nitrosodimethylamine	<5	ug/L	EPA 625	10/11/11	KTL	
N-Nitrosodiphenylamine	<5	ug/L	EPA 625	10/11/11	KTL	
Naphthalene by Method 625	<5	ug/L	EPA 625	10/11/11	KTL	
Nitrobenzene	<5	ug/L	EPA 625	10/11/11	KTL	
Pentachlorophenol	<5	ug/L	EPA 625	10/11/11	KTL	
Phenanthrene	<5	ug/L	EPA 625	10/11/11	KTL	
Phenol	<5	ug/L	EPA 625	10/11/11	KTL	
Pyrene	<5	ug/L	EPA 625	10/11/11	KTL	

Sample ID : ***"Plant Effluent, Grab"***

Sampled By : *AV of City of Cadillac, Water Resources*

Sample Date : *10/03/11*

Sample Time : *0830*

Date Received : *10/04/11*

Sample Type : *aqueous*

KAR Sample No. : *114264-02*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Phenols, total	<0.02	mg/L	EPA 420.4	10/17/11	ALK	
Prior. Poll. volatiles	See below		EPA 624	10/10/11	JAR	
Prep, VOA	Completed		EPA 624	10/10/11	JAR	
1,1,1-Trichloroethane	<1	ug/L	EPA 624	10/10/11	JAR	
1,1,2,2-Tetrachloroethane	<1	ug/L	EPA 624	10/10/11	JAR	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *114264*

Date Reported : *10/18/11*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: *Analysis of two aqueous samples for NPDES permit.*

Sample ID : ***"Plant Effluent, Grab"***

Sampled By : *AV of City of Cadillac, Water Resources*

Sample Date : *10/03/11*

Sample Time : *0830*

Date Received : *10/04/11*

Sample Type : *aqueous*

KAR Sample No. : *114264-02*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,1,2-Trichloroethane	<1	ug/L	EPA 624	10/10/11	JAR	
1,1-Dichloroethane	<1	ug/L	EPA 624	10/10/11	JAR	
1,1-Dichloroethene	<1	ug/L	EPA 624	10/10/11	JAR	
1,2-Dichlorobenzene	<1	ug/L	EPA 624	10/10/11	JAR	
1,2-Dichloroethane	<1	ug/L	EPA 624	10/10/11	JAR	
1,2-Dichloropropane	<1	ug/L	EPA 624	10/10/11	JAR	
1,3-Dichlorobenzene	<1	ug/L	EPA 624	10/10/11	JAR	
1,4-Dichlorobenzene	<1	ug/L	EPA 624	10/10/11	JAR	
2-Chloroethylvinyl ether	<10	ug/L	EPA 624	10/10/11	JAR	
Acrolein	<20	ug/L	EPA 624	10/10/11	JAR	
Acrylonitrile	<2	ug/L	EPA 624	10/10/11	JAR	
Benzene	<1	ug/L	EPA 624	10/10/11	JAR	
Bromodichloromethane	<1	ug/L	EPA 624	10/10/11	JAR	
Bromoform	<1	ug/L	EPA 624	10/10/11	JAR	
Bromomethane	<1	ug/L	EPA 624	10/10/11	JAR	
Carbon tetrachloride	<1	ug/L	EPA 624	10/10/11	JAR	
Chlorobenzene	<1	ug/L	EPA 624	10/10/11	JAR	
Chloroethane	<1	ug/L	EPA 624	10/10/11	JAR	
Chloroform	<1	ug/L	EPA 624	10/10/11	JAR	
Chloromethane	<1	ug/L	EPA 624	10/10/11	JAR	
Cis-1,3-Dichloropropene	<1	ug/L	EPA 624	10/10/11	JAR	
Dibromochloromethane	<1	ug/L	EPA 624	10/10/11	JAR	
Ethylbenzene	<1	ug/L	EPA 624	10/10/11	JAR	
Methylene chloride	<1	ug/L	EPA 624	10/10/11	JAR	
Tetrachloroethene	<1	ug/L	EPA 624	10/10/11	JAR	
Toluene	<1	ug/L	EPA 624	10/10/11	JAR	
Trans-1,2-Dichloroethene	<1	ug/L	EPA 624	10/10/11	JAR	
Trans-1,3-Dichloropropene	<1	ug/L	EPA 624	10/10/11	JAR	
Trichloroethene	<1	ug/L	EPA 624	10/10/11	JAR	
Trichlorofluoromethane	<1	ug/L	EPA 624	10/10/11	JAR	
Vinyl chloride	<1	ug/L	EPA 624	10/10/11	JAR	

KAR Laboratories, Inc. (269-381-9666) **Chain-of-Custody** **www.karlabs.com**

Client: Cadillac WWTP
800 N. Lake St
Cadillac, MI 49601

Attn: Amy Vail 2368

Phone: (231) 775-2876 x1031 Fax: (231) 775-2876

Cell: ( )

Email: lab@cadillac-mi.net

Project Name: Additional NPDES Monitoring

Project # /WSSN#

P.O. #

Sampled by: ☐ KAR ☑ Client Initials AV

Turnaround Time:
☑ 10-business days
☐ 5-business days (x 1.5)
☐ 3-business days (x 2.0)
☐ Emergency (quote)
☐ Monthly

Samples rec'd
Cold: (Y) N Intact: (Y) N

KAR use only
Proj#: 114264
Login: HES
Date: 10/4/11
c-of-c (Y) N
labels (Y) N
LODs Y (N)
contain (Y) N
headspace Y (N)
amount (Y) N
hold time (Y) N
rush cost Y (N)
FRB Y (N)

#	Sample Identification	Date	Time	Matrix	Type	Size	#	VOC	Phenols	Acid/Base Extractables	Acid/Base Extractables	Hardness	Metals				Remarks
1	Effluent	10/3/11	8:30a	Aqueous	Glass	40mL	2	X									-02
2	"	"	"	"	Amber Glass	1L	1		X	(H2SO4)							↓
3	"	"	7:45a	"	Amber Glass	1L	1			X							-01
4	"	"	"	"	Amber Glass	1L	1				X						
5	"	"	"	"	Plastic	125mL	1					X					
	"	"	"	"	Plastic	250mL	1						X	(HNO_3)			↓

Relinquished By: AV 1NS 2PM
Received By: Sandy Motz
Date/Time: 10-4-11 11:35am

Relinquished By:
Received By:
Date/Time:

Notes: See Attached Sheet for specifics 1 Amber Broken

Form KC020.F1.02 Effective Date: December 29, 2009

4425 Manchester Road
Kalamazoo, MI 49001
Phone 269 381-9666
Fax 269 381-9698
www.karlabs.com

City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : ***Ms. Amy Vail***

KAR Project No. :	***110856***
Date Reported :	***03/16/11***
Date Activated :	***03/02/11***
Date Due :	***03/16/11***
Date Validated :	***03/16/11***

Project Description : ***Analysis of two aqueous samples for NPDES permit.***

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 110856. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

David R. Alkema

David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: ***City of Cadillac, Water Resources***

KAR Project No. : ***110856***

Date Reported : ***03/16/11***

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Composite"***

Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *03/01/11*
Sample Time : *0738*

Date Received : *03/02/11*
Sample Type : *aqueous*
KAR Sample No. : ***110856-01***

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, metals	Completed		EPA 30xx,200.x	03/03/11	PML	
Antimony, total, low level	<1	ug/L	EPA 200.8	03/11/11	PML	
Arsenic, total, low level	2	ug/L	EPA 200.8	03/15/11	PML	
Barium, total, low level	15	ug/L	EPA 200.8	03/11/11	PML	
Beryllium, total, low level	<1	ug/L	EPA 200.8	03/11/11	PML	
Boron, total, low level	182	ug/L	EPA 200.7	03/15/11	DBL	
Cadmium, total, low level	<0.2	ug/L	EPA 200.8	03/15/11	PML	
Chromium, total, low level	<5	ug/L	EPA 200.8	03/11/11	PML	
Copper, total, low level	15	ug/L	EPA 200.8	03/11/11	PML	
Lead, total, low level	<1	ug/L	EPA 200.8	03/11/11	PML	
Nickel, total, low level	<5	ug/L	EPA 200.8	03/11/11	PML	
Silver, total, low level	<0.5	ug/L	EPA 200.8	03/15/11	PML	
Thallium, total, low level	<1	ug/L	EPA 200.8	03/11/11	PML	
Hardness	223	mg/L (as CaCO3)	EPA 130.2	03/11/11	PML	
Prior. Poll. BN by 625	See below		EPA 625	03/08/11	KTL	
Prior. Poll. acids by 625	See below		EPA 625	03/08/11	KTL	
Prep, SV Acid/BN	Completed		EPA 625	03/07/11	KTL	
1,2,4-Trichlorobenzene 625	<5	ug/L	EPA 625	03/08/11	KTL	
1,2-Diphenylhydrazine	<5	ug/L	EPA 625	03/08/11	KTL	
2,4,6-Trichlorophenol	<5	ug/L	EPA 625	03/08/11	KTL	
2,4-Dichlorophenol	<5	ug/L	EPA 625	03/08/11	KTL	
2,4-Dimethylphenol	<5	ug/L	EPA 625	03/08/11	KTL	
2,4-Dinitrophenol	<20	ug/L	EPA 625	03/08/11	KTL	
2,4-Dinitrotoluene	<5	ug/L	EPA 625	03/08/11	KTL	
2,6-Dinitrotoluene	<5	ug/L	EPA 625	03/08/11	KTL	
2-Chloronaphthalene	<5	ug/L	EPA 625	03/08/11	KTL	
2-Chlorophenol	<5	ug/L	EPA 625	03/08/11	KTL	
2-Methyl-4,6-dinitrophenol	<20	ug/L	EPA 625	03/08/11	KTL	
2-Nitrophenol	<5	ug/L	EPA 625	03/08/11	KTL	
3,3'-Dichlorobenzidine	<20	ug/L	EPA 625	03/08/11	KTL	
4-Bromophenyl phenyl ether	<5	ug/L	EPA 625	03/08/11	KTL	
4-Chloro-3-methylphenol	<5	ug/L	EPA 625	03/08/11	KTL	
4-Chlorophenyl phenyl ether	<5	ug/L	EPA 625	03/08/11	KTL	
4-Nitrophenol	<20	ug/L	EPA 625	03/08/11	KTL	
Acenaphthene	<5	ug/L	EPA 625	03/08/11	KTL	
Acenaphthylene	<5	ug/L	EPA 625	03/08/11	KTL	
Anthracene	<5	ug/L	EPA 625	03/08/11	KTL	
Benzidine	<50	ug/L	EPA 625	03/08/11	KTL	
Benzo(a)anthracene	<5	ug/L	EPA 625	03/08/11	KTL	
Benzo(a)pyrene	<5	ug/L	EPA 625	03/08/11	KTL	
Benzo(b)fluoranthene	<5	ug/L	EPA 625	03/08/11	KTL	
Benzo(ghi)perylene	<5	ug/L	EPA 625	03/08/11	KTL	
Benzo(k)fluoranthene	<5	ug/L	EPA 625	03/08/11	KTL	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *110856*

Date Reported : *03/16/11*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: *Analysis of two aqueous samples for NPDES permit.*

Sample ID : ***"Plant Effluent, Composite"***

Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *03/01/11*
Sample Time : *0738*

Date Received : *03/02/11*
Sample Type : *aqueous*
KAR Sample No. : *110856-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Bis(2-chloroethoxy)methane	<5	ug/L	EPA 625	03/08/11	KTL	
Bis(2-chloroethyl)ether	<5	ug/L	EPA 625	03/08/11	KTL	
Bis(2-chloroisopropyl)ether	<5	ug/L	EPA 625	03/08/11	KTL	
Bis(2-ethylhexyl)phthalate	<5	ug/L	EPA 625	03/08/11	KTL	
Butylbenzyl phthalate	<5	ug/L	EPA 625	03/08/11	KTL	
Chrysene	<5	ug/L	EPA 625	03/08/11	KTL	
Di-n-butyl phthalate	<5	ug/L	EPA 625	03/08/11	KTL	
Di-n-octyl phthalate	<5	ug/L	EPA 625	03/08/11	KTL	
Dibenzo(ah)anthracene	<5	ug/L	EPA 625	03/08/11	KTL	
Diethyl phthalate	<5	ug/L	EPA 625	03/08/11	KTL	
Dimethyl phthalate	<5	ug/L	EPA 625	03/08/11	KTL	
Fluoranthene	<5	ug/L	EPA 625	03/08/11	KTL	
Fluorene	<5	ug/L	EPA 625	03/08/11	KTL	
Hexachlorobenzene	<5	ug/L	EPA 625	03/08/11	KTL	
Hexachlorobutadiene	<5	ug/L	EPA 625	03/08/11	KTL	
Hexachlorocyclopentadiene	<5	ug/L	EPA 625	03/08/11	KTL	
Hexachloroethane	<5	ug/L	EPA 625	03/08/11	KTL	
Indeno(123cd)pyrene	<5	ug/L	EPA 625	03/08/11	KTL	
Isophorone	<5	ug/L	EPA 625	03/08/11	KTL	
N-Nitrosodi-n-propylamine	<5	ug/L	EPA 625	03/08/11	KTL	
N-Nitrosodimethylamine	<5	ug/L	EPA 625	03/08/11	KTL	
N-Nitrosodiphenylamine	<5	ug/L	EPA 625	03/08/11	KTL	
Naphthalene by Method 625	<5	ug/L	EPA 625	03/08/11	KTL	
Nitrobenzene	<5	ug/L	EPA 625	03/08/11	KTL	
Pentachlorophenol	<5	ug/L	EPA 625	03/08/11	KTL	
Phenanthrene	<5	ug/L	EPA 625	03/08/11	KTL	
Phenol	<5	ug/L	EPA 625	03/08/11	KTL	
Pyrene	<5	ug/L	EPA 625	03/08/11	KTL	

Sample ID : ***"Plant Effluent, Grab"***

Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *03/01/11*
Sample Time : *0740*

Date Received : *03/02/11*
Sample Type : *aqueous*
KAR Sample No. : *110856-02*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Phenols, total	<0.02	mg/L	EPA 420.4	03/04/11	ALK	
Prior. Poll. volatiles	See below		EPA 624	03/07/11	JAR	
Prep, VOA	Completed		EPA 624	03/07/11	JAR	
1,1,1-Trichloroethane	<1	ug/L	EPA 624	03/07/11	JAR	
1,1,2,2-Tetrachloroethane	<1	ug/L	EPA 624	03/07/11	JAR	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *110856*

Date Reported : *03/16/11*

Attest: *David R. Alkema*
David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Grab"***

Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *03/01/11*
Sample Time : *0740*

Date Received : *03/02/11*
Sample Type : *aqueous*
KAR Sample No. : *110856-02*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,1,2-Trichloroethane	<1	ug/L	EPA 624	03/07/11	JAR	
1,1-Dichloroethane	<1	ug/L	EPA 624	03/07/11	JAR	
1,1-Dichloroethene	<1	ug/L	EPA 624	03/07/11	JAR	
1,2-Dichlorobenzene	<1	ug/L	EPA 624	03/07/11	JAR	
1,2-Dichloroethane	<1	ug/L	EPA 624	03/07/11	JAR	
1,2-Dichloropropane	<1	ug/L	EPA 624	03/07/11	JAR	
1,3-Dichlorobenzene	<1	ug/L	EPA 624	03/07/11	JAR	
1,4-Dichlorobenzene	<1	ug/L	EPA 624	03/07/11	JAR	
2-Chloroethylvinyl ether	<10	ug/L	EPA 624	03/07/11	JAR	
Acrolein	<20	ug/L	EPA 624	03/07/11	JAR	
Acrylonitrile	<2	ug/L	EPA 624	03/07/11	JAR	
Benzene	<1	ug/L	EPA 624	03/07/11	JAR	
Bromodichloromethane	<1	ug/L	EPA 624	03/07/11	JAR	
Bromoform	<1	ug/L	EPA 624	03/07/11	JAR	
Bromomethane	<1	ug/L	EPA 624	03/07/11	JAR	
Carbon tetrachloride	<1	ug/L	EPA 624	03/07/11	JAR	
Chlorobenzene	<1	ug/L	EPA 624	03/07/11	JAR	
Chloroethane	<1	ug/L	EPA 624	03/07/11	JAR	
Chloroform	<1	ug/L	EPA 624	03/07/11	JAR	
Chloromethane	<1	ug/L	EPA 624	03/07/11	JAR	
Cis-1,3-Dichloropropene	<1	ug/L	EPA 624	03/07/11	JAR	
Dibromochloromethane	<1	ug/L	EPA 624	03/07/11	JAR	
Ethylbenzene	<1	ug/L	EPA 624	03/07/11	JAR	
Methylene chloride	<1	ug/L	EPA 624	03/07/11	JAR	
Tetrachloroethene	<1	ug/L	EPA 624	03/07/11	JAR	
Toluene	<1	ug/L	EPA 624	03/07/11	JAR	
Trans-1,2-Dichloroethene	<1	ug/L	EPA 624	03/07/11	JAR	
Trans-1,3-Dichloropropene	<1	ug/L	EPA 624	03/07/11	JAR	
Trichloroethene	<1	ug/L	EPA 624	03/07/11	JAR	
Trichlorofluoromethane	<1	ug/L	EPA 624	03/07/11	JAR	
Vinyl chloride	<1	ug/L	EPA 624	03/07/11	JAR	

4425 Manchester Road
Kalamazoo, MI 49001
Phone 269 381-9666
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www.karlabs.com

City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : ***Ms. Amy Vail***

KAR Project No. :	***101697***
Date Reported :	***05/19/10***
Date Activated :	***05/05/10***
Date Due :	***05/19/10***
Date Validated :	***05/19/10***

Project Description : ***Analysis of two aqueous samples for NPDES permit.***

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 101697. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

David R. Alkema

David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *101697*

Date Reported : *05/19/10*

Attest: David R. Alkema

David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Composite"***

Sampled By : *AV of City of Cadillac, Water Resources*

Sample Date : *05/04/10*

Sample Time : *0730*

Date Received : *05/05/10*

Sample Type : *aqueous*

KAR Sample No. : *101697-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, metals	Completed		EPA 30xx,200.x	05/06/10	DBL	
Antimony, total, low level	<1	ug/L	EPA 200.8	05/18/10	PML	
Arsenic, total, low level	<1	ug/L	EPA 200.8	05/18/10	PML	
Barium, total, low level	18	ug/L	EPA 200.8	05/18/10	PML	
Beryllium, total, low level	<1	ug/L	EPA 200.8	05/18/10	PML	
Boron, total, low level	220	ug/L	EPA 200.7	05/11/10	DBL	
Cadmium, total, low level	<0.2	ug/L	EPA 200.8	05/18/10	PML	
Calcium, total	61.5	mg/L	EPA 200.7	05/07/10	DBL	
Chromium, total, low level	<5	ug/L	EPA 200.8	05/18/10	PML	
Copper, total, low level	8	ug/L	EPA 200.8	05/18/10	PML	
Lead, total, low level	<1	ug/L	EPA 200.8	05/18/10	PML	
Magnesium, total	17.1	mg/L	EPA 200.7	05/07/10	DBL	
Nickel, total, low level	3	ug/L	EPA 200.8	05/18/10	PML	
Silver, total, low level	<0.5	ug/L	EPA 200.8	05/18/10	PML	
Thallium, total, low level	<2	ug/L	EPA 200.8	05/18/10	PML	
Hardness	224	mg/L (as CaCO3)	EPA 130.2	05/07/10	PML	
Prior. Poll. BN by 625	See below		EPA 625	05/13/10	KTL	
Prior. Poll. acids by 625	See below		EPA 625	05/13/10	KTL	
Prep, SV Acid/BN	Completed		EPA 625	05/06/10	GMB	
1,2,4-Trichlorobenzene 625	<5	ug/L	EPA 625	05/13/10	KTL	
1,2-Diphenylhydrazine	<5	ug/L	EPA 625	05/13/10	KTL	
2,4,6-Trichlorophenol	<5	ug/L	EPA 625	05/13/10	KTL	
2,4-Dichlorophenol	<5	ug/L	EPA 625	05/13/10	KTL	
2,4-Dimethylphenol	<5	ug/L	EPA 625	05/13/10	KTL	
2,4-Dinitrophenol	<20	ug/L	EPA 625	05/13/10	KTL	
2,4-Dinitrotoluene	<5	ug/L	EPA 625	05/13/10	KTL	
2,6-Dinitrotoluene	<5	ug/L	EPA 625	05/13/10	KTL	
2-Chloronaphthalene	<5	ug/L	EPA 625	05/13/10	KTL	
2-Chlorophenol	<5	ug/L	EPA 625	05/13/10	KTL	
2-Methyl-4,6-dinitrophenol	<20	ug/L	EPA 625	05/13/10	KTL	
2-Nitrophenol	<5	ug/L	EPA 625	05/13/10	KTL	
3,3'-Dichlorobenzidine	<20	ug/L	EPA 625	05/13/10	KTL	
4-Bromophenyl phenyl ether	<5	ug/L	EPA 625	05/13/10	KTL	
4-Chloro-3-methylphenol	<5	ug/L	EPA 625	05/13/10	KTL	
4-Chlorophenyl phenyl ether	<5	ug/L	EPA 625	05/13/10	KTL	
4-Nitrophenol	<20	ug/L	EPA 625	05/13/10	KTL	
Acenaphthene	<5	ug/L	EPA 625	05/13/10	KTL	
Acenaphthylene	<5	ug/L	EPA 625	05/13/10	KTL	
Anthracene	<5	ug/L	EPA 625	05/13/10	KTL	
Benzidine	<50	ug/L	EPA 625	05/13/10	KTL	
Benzo(a)anthracene	<5	ug/L	EPA 625	05/13/10	KTL	
Benzo(a)pyrene	<5	ug/L	EPA 625	05/13/10	KTL	
Benzo(b)fluoranthene	<5	ug/L	EPA 625	05/13/10	KTL	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *101697*
Date Reported : *05/19/10*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Composite"***
Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *05/04/10*
Sample Time : *0730*

Date Received : *05/05/10*
Sample Type : *aqueous*
KAR Sample No. : *101697-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Benzo(ghi)perylene	<5	ug/L	EPA 625	05/13/10	KTL	
Benzo(k)fluoranthene	<5	ug/L	EPA 625	05/13/10	KTL	
Bis(2-chloroethoxy)methane	<5	ug/L	EPA 625	05/13/10	KTL	
Bis(2-chloroethyl)ether	<5	ug/L	EPA 625	05/13/10	KTL	
Bis(2-chloroisopropyl)ether	<5	ug/L	EPA 625	05/13/10	KTL	
Bis(2-ethylhexyl)phthalate	<5	ug/L	EPA 625	05/13/10	KTL	
Butylbenzyl phthalate	<5	ug/L	EPA 625	05/13/10	KTL	
Chrysene	<5	ug/L	EPA 625	05/13/10	KTL	
Di-n-butyl phthalate	<5	ug/L	EPA 625	05/13/10	KTL	
Di-n-octyl phthalate	<5	ug/L	EPA 625	05/13/10	KTL	
Dibenzo(ah)anthracene	<5	ug/L	EPA 625	05/13/10	KTL	
Diethyl phthalate	<5	ug/L	EPA 625	05/13/10	KTL	
Dimethyl phthalate	<5	ug/L	EPA 625	05/13/10	KTL	
Fluoranthene	<5	ug/L	EPA 625	05/13/10	KTL	
Fluorene	<5	ug/L	EPA 625	05/13/10	KTL	
Hexachlorobenzene	<5	ug/L	EPA 625	05/13/10	KTL	
Hexachlorobutadiene	<5	ug/L	EPA 625	05/13/10	KTL	
Hexachlorocyclopentadiene	<5	ug/L	EPA 625	05/13/10	KTL	
Hexachloroethane	<5	ug/L	EPA 625	05/13/10	KTL	
Indeno(123cd)pyrene	<5	ug/L	EPA 625	05/13/10	KTL	
Isophorone	<5	ug/L	EPA 625	05/13/10	KTL	
N-Nitrosodi-n-propylamine	<5	ug/L	EPA 625	05/13/10	KTL	
N-Nitrosodimethylamine	<5	ug/L	EPA 625	05/13/10	KTL	
N-Nitrosodiphenylamine	<5	ug/L	EPA 625	05/13/10	KTL	
Naphthalene by Method 625	<5	ug/L	EPA 625	05/13/10	KTL	
Nitrobenzene	<5	ug/L	EPA 625	05/13/10	KTL	
Pentachlorophenol	<5	ug/L	EPA 625	05/13/10	KTL	
Phenanthrene	<5	ug/L	EPA 625	05/13/10	KTL	
Phenol	<5	ug/L	EPA 625	05/13/10	KTL	
Pyrene	<5	ug/L	EPA 625	05/13/10	KTL	

Sample ID : ***"Plant Effluent, Grab"***
Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *05/04/10*
Sample Time : *0722*

Date Received : *05/05/10*
Sample Type : *aqueous*
KAR Sample No. : *101697-02*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Phenols, total	<0.02	mg/L	EPA 420.4	05/06/10	MTW	
Prior. Poll. volatiles	See below		EPA 624	05/11/10	JAR	
Prep, VOA	Completed		EPA 624	05/11/10	JAR	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *101697*

Date Reported : *05/19/10*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Grab"***

Sampled By : *AV of City of Cadillac, Water Resources*

Sample Date : *05/04/10*

Sample Time : *0722*

Date Received : *05/05/10*

Sample Type : *aqueous*

KAR Sample No. : *101697-02*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,1,1-Trichloroethane	<1	ug/L	EPA 624	05/11/10	JAR	
1,1,2,2-Tetrachloroethane	<1	ug/L	EPA 624	05/11/10	JAR	
1,1,2-Trichloroethane	<1	ug/L	EPA 624	05/11/10	JAR	
1,1-Dichloroethane	<1	ug/L	EPA 624	05/11/10	JAR	
1,1-Dichloroethene	<1	ug/L	EPA 624	05/11/10	JAR	
1,2-Dichlorobenzene	<1	ug/L	EPA 624	05/11/10	JAR	
1,2-Dichloroethane	<1	ug/L	EPA 624	05/11/10	JAR	
1,2-Dichloropropane	<1	ug/L	EPA 624	05/11/10	JAR	
1,3-Dichlorobenzene	<1	ug/L	EPA 624	05/11/10	JAR	
1,4-Dichlorobenzene	<1	ug/L	EPA 624	05/11/10	JAR	
2-Chloroethylvinyl ether	<10	ug/L	EPA 624	05/11/10	JAR	
Acrolein	<20	ug/L	EPA 624	05/11/10	JAR	
Acrylonitrile	<2	ug/L	EPA 624	05/11/10	JAR	
Benzene	<1	ug/L	EPA 624	05/11/10	JAR	
Bromodichloromethane	<1	ug/L	EPA 624	05/11/10	JAR	
Bromoform	<1	ug/L	EPA 624	05/11/10	JAR	
Bromomethane	<1	ug/L	EPA 624	05/11/10	JAR	
Carbon tetrachloride	<1	ug/L	EPA 624	05/11/10	JAR	
Chlorobenzene	<1	ug/L	EPA 624	05/11/10	JAR	
Chloroethane	<1	ug/L	EPA 624	05/11/10	JAR	
Chloroform	<1	ug/L	EPA 624	05/11/10	JAR	
Chloromethane	<1	ug/L	EPA 624	05/11/10	JAR	
Cis-1,3-Dichloropropene	<1	ug/L	EPA 624	05/11/10	JAR	
Dibromochloromethane	<1	ug/L	EPA 624	05/11/10	JAR	
Ethylbenzene	<1	ug/L	EPA 624	05/11/10	JAR	
Methylene chloride	<1	ug/L	EPA 624	05/11/10	JAR	
Tetrachloroethene	<1	ug/L	EPA 624	05/11/10	JAR	
Toluene	<1	ug/L	EPA 624	05/11/10	JAR	
Trans-1,2-Dichloroethene	<1	ug/L	EPA 624	05/11/10	JAR	
Trans-1,3-Dichloropropene	<1	ug/L	EPA 624	05/11/10	JAR	
Trichloroethene	<1	ug/L	EPA 624	05/11/10	JAR	
Trichlorofluoromethane	<1	ug/L	EPA 624	05/11/10	JAR	
Vinyl chloride	<1	ug/L	EPA 624	05/11/10	JAR	

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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : ***Mr. Jeff Dietlin***

KAR Project No. :	***093114***
Date Reported :	***08/27/09***
Date Activated :	***08/13/09***
Date Due :	***08/27/09***
Date Validated :	***08/27/09***

Project Description : ***Analysis of two aqueous samples for NPDES permit.***

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 093114. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

David R. Alkema

David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *093114*

Date Reported : *08/27/09*

Attest: David R. Alkema

David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Composite"***

Sampled By : *AJV of City of Cadillac, Water Resources*

Sample Date : *08/12/09*

Sample Time : *0716*

Date Received : *08/13/09*

Sample Type : *aqueous*

KAR Sample No. : *093114-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, metals	Completed		EPA 30xx,200.x	08/14/09	DBL	
Antimony, total, low level	<1	ug/L	EPA 200.8	08/20/09	PML	
Arsenic, total, low level	2	ug/L	EPA 200.8	08/20/09	PML	
Barium, total, low level	16	ug/L	EPA 200.8	08/20/09	PML	
Beryllium, total, low level	<1	ug/L	EPA 200.8	08/20/09	PML	
Boron, total, low level	355	ug/L	EPA 200.7	08/17/09	DBL	
Cadmium, total, low level	<0.2	ug/L	EPA 200.8	08/20/09	PML	
Calcium, total	77.6	mg/L	EPA 200.7	08/17/09	DBL	
Chromium, total, low level	<5	ug/L	EPA 200.8	08/20/09	PML	
Copper, total, low level	4	ug/L	EPA 200.8	08/20/09	PML	
Lead, total, low level	<1	ug/L	EPA 200.8	08/20/09	PML	
Magnesium, total	18.4	mg/L	EPA 200.7	08/17/09	DBL	
Nickel, total, low level	5	ug/L	EPA 200.8	08/20/09	PML	
Silver, total, low level	<0.5	ug/L	EPA 200.8	08/20/09	PML	
Thallium, total, low level	<2	ug/L	EPA 200.8	08/20/09	PML	
Hardness	270	mg/L (as CaCO3)	SM 2340 B	08/17/09	PML	
Prior. Poll. acids	See below		EPA 625	08/17/09	KTL	
Prior. Poll. base-neutrals	See below		EPA 625	08/17/09	KTL	
Prep, SV Acid/BN	Completed		EPA 625	08/17/09	GMB	
1,2,4-Trichlorobenzene 8270	<5	ug/L	EPA 625	08/17/09	KTL	
1,2-Dichlorobenzene by 8270	<5	ug/L	EPA 625	08/17/09	KTL	
1,2-Diphenylhydrazine	<5	ug/L	EPA 625	08/17/09	KTL	
1,3-Dichlorobenzene by 8270	<5	ug/L	EPA 625	08/17/09	KTL	
1,4-Dichlorobenzene by 8270	<5	ug/L	EPA 625	08/17/09	KTL	
2,4,6-Trichlorophenol	<5	ug/L	EPA 625	08/17/09	KTL	
2,4-Dichlorophenol	<5	ug/L	EPA 625	08/17/09	KTL	
2,4-Dimethylphenol	<5	ug/L	EPA 625	08/17/09	KTL	
2,4-Dinitrophenol	<20	ug/L	EPA 625	08/17/09	KTL	
2,4-Dinitrotoluene	<5	ug/L	EPA 625	08/17/09	KTL	
2,6-Dinitrotoluene	<5	ug/L	EPA 625	08/17/09	KTL	
2-Chloronaphthalene	<5	ug/L	EPA 625	08/17/09	KTL	
2-Chlorophenol	<5	ug/L	EPA 625	08/17/09	KTL	
2-Methyl-4,6-dinitrophenol	<20	ug/L	EPA 625	08/17/09	KTL	
2-Nitrophenol	<5	ug/L	EPA 625	08/17/09	KTL	
3,3'-Dichlorobenzidine	<20	ug/L	EPA 625	08/17/09	KTL	
4-Bromophenyl phenyl ether	<5	ug/L	EPA 625	08/17/09	KTL	
4-Chloro-3-methylphenol	<5	ug/L	EPA 625	08/17/09	KTL	
4-Chlorophenyl phenyl ether	<5	ug/L	EPA 625	08/17/09	KTL	
4-Nitrophenol	<20	ug/L	EPA 625	08/17/09	KTL	
Acenaphthene	<5	ug/L	EPA 625	08/17/09	KTL	
Acenaphthylene	<5	ug/L	EPA 625	08/17/09	KTL	
Anthracene	<5	ug/L	EPA 625	08/17/09	KTL	
Benzidine	<50	ug/L	EPA 625	08/17/09	KTL	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *093114*

Date Reported : *08/27/09*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Composite"***

Sampled By : *AJV of City of Cadillac, Water Resources*
Sample Date : *08/12/09*
Sample Time : *0716*

Date Received : *08/13/09*
Sample Type : *aqueous*
KAR Sample No. : *093114-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Benzo(a)anthracene	<5	ug/L	EPA 625	08/17/09	KTL	
Benzo(a)pyrene	<5	ug/L	EPA 625	08/17/09	KTL	
Benzo(b)fluoranthene	<5	ug/L	EPA 625	08/17/09	KTL	
Benzo(ghi)perylene	<5	ug/L	EPA 625	08/17/09	KTL	
Benzo(k)fluoranthene	<5	ug/L	EPA 625	08/17/09	KTL	
Bis(2-chloroethoxy)methane	<5	ug/L	EPA 625	08/17/09	KTL	
Bis(2-chloroethyl)ether	<5	ug/L	EPA 625	08/17/09	KTL	
Bis(2-chloroisopropyl)ether	<5	ug/L	EPA 625	08/17/09	KTL	
Bis(2-ethylhexyl)phthalate	<5	ug/L	EPA 625	08/17/09	KTL	
Butylbenzyl phthalate	<5	ug/L	EPA 625	08/17/09	KTL	
Chrysene	<5	ug/L	EPA 625	08/17/09	KTL	
Di-n-butyl phthalate	<5	ug/L	EPA 625	08/17/09	KTL	
Di-n-octyl phthalate	<5	ug/L	EPA 625	08/17/09	KTL	
Dibenzo(ah)anthracene	<5	ug/L	EPA 625	08/17/09	KTL	
Diethyl phthalate	<5	ug/L	EPA 625	08/17/09	KTL	
Dimethyl phthalate	<5	ug/L	EPA 625	08/17/09	KTL	
Fluoranthene	<5	ug/L	EPA 625	08/17/09	KTL	
Fluorene	<5	ug/L	EPA 625	08/17/09	KTL	
Hexachlorobenzene	<5	ug/L	EPA 625	08/17/09	KTL	
Hexachlorobutadiene	<5	ug/L	EPA 625	08/17/09	KTL	
Hexachlorocyclopentadiene	<5	ug/L	EPA 625	08/17/09	KTL	
Hexachloroethane	<5	ug/L	EPA 625	08/17/09	KTL	
Indeno(123cd)pyrene	<5	ug/L	EPA 625	08/17/09	KTL	
Isophorone	<5	ug/L	EPA 625	08/17/09	KTL	
N-Nitrosodi-n-propylamine	<5	ug/L	EPA 625	08/17/09	KTL	
N-Nitrosodimethylamine	<5	ug/L	EPA 625	08/17/09	KTL	
N-Nitrosodiphenylamine	<5	ug/L	EPA 625	08/17/09	KTL	
Naphthalene by Method 8270	<5	ug/L	EPA 625	08/17/09	KTL	
Nitrobenzene	<5	ug/L	EPA 625	08/17/09	KTL	
Pentachlorophenol	<5	ug/L	EPA 625	08/17/09	KTL	
Phenanthrene	<5	ug/L	EPA 625	08/17/09	KTL	
Phenol	<5	ug/L	EPA 625	08/17/09	KTL	
Pyrene	<5	ug/L	EPA 625	08/17/09	KTL	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : ***093114***

Date Reported : ***08/27/09***

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: ***Analysis of two aqueous samples for NPDES permit.***

Sample ID : ***"Plant Effluent, Grab"***

Sampled By : *AJV of City of Cadillac, Water Resources*
Sample Date : *08/12/09*
Sample Time : *0722*

Date Received : *08/13/09*
Sample Type : *aqueous*
KAR Sample No. : ***093114-02***

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Phenols, total	<0.02	mg/L	EPA 420.4	08/17/09	MTW	
Prior. Poll. volatiles	See below		EPA 624	08/24/09	JAR	
Prep, VOA	Completed		EPA 624	08/24/09	JAR	
1,1,1-Trichloroethane	<1	ug/L	EPA 624	08/24/09	JAR	
1,1,2,2-Tetrachloroethane	<1	ug/L	EPA 624	08/24/09	JAR	
1,1,2-Trichloroethane	<1	ug/L	EPA 624	08/24/09	JAR	
1,1-Dichloroethane	<1	ug/L	EPA 624	08/24/09	JAR	
1,1-Dichloroethene	<1	ug/L	EPA 624	08/24/09	JAR	
1,2-Dichlorobenzene	<1	ug/L	EPA 624	08/24/09	JAR	
1,2-Dichloroethane	<1	ug/L	EPA 624	08/24/09	JAR	
1,2-Dichloropropane	<1	ug/L	EPA 624	08/24/09	JAR	
1,3-Dichlorobenzene	<1	ug/L	EPA 624	08/24/09	JAR	
1,4-Dichlorobenzene	<1	ug/L	EPA 624	08/24/09	JAR	
2-Chloroethylvinyl ether	<10	ug/L	EPA 624	08/24/09	JAR	
Acrolein	<20	ug/L	EPA 624	08/24/09	JAR	
Acrylonitrile	<2	ug/L	EPA 624	08/24/09	JAR	
Benzene	<1	ug/L	EPA 624	08/24/09	JAR	
Bromodichloromethane	<1	ug/L	EPA 624	08/24/09	JAR	
Bromoform	<1	ug/L	EPA 624	08/24/09	JAR	
Bromomethane	<1	ug/L	EPA 624	08/24/09	JAR	
Carbon tetrachloride	<1	ug/L	EPA 624	08/24/09	JAR	
Chlorobenzene	<1	ug/L	EPA 624	08/24/09	JAR	
Chloroethane	<1	ug/L	EPA 624	08/24/09	JAR	
Chloroform	<1	ug/L	EPA 624	08/24/09	JAR	
Chloromethane	<1	ug/L	EPA 624	08/24/09	JAR	
Cis-1,3-Dichloropropene	<1	ug/L	EPA 624	08/24/09	JAR	
Dibromochloromethane	<1	ug/L	EPA 624	08/24/09	JAR	
Ethylbenzene	<1	ug/L	EPA 624	08/24/09	JAR	
Methylene chloride	<1	ug/L	EPA 624	08/24/09	JAR	
Tetrachloroethene	<1	ug/L	EPA 624	08/24/09	JAR	
Toluene	<1	ug/L	EPA 624	08/24/09	JAR	
Trans-1,2-Dichloroethene	<1	ug/L	EPA 624	08/24/09	JAR	
Trans-1,3-Dichloropropene	<1	ug/L	EPA 624	08/24/09	JAR	
Trichloroethene	<1	ug/L	EPA 624	08/24/09	JAR	
Trichlorofluoromethane	<1	ug/L	EPA 624	08/24/09	JAR	
Vinyl chloride	<1	ug/L	EPA 624	08/24/09	JAR	

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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : ***Ms. Amy Vail***

KAR Project No. :	***114817***
Date Reported :	***11/10/11***
Date Activated :	***11/02/11***
Date Due :	***11/16/11***
Date Validated :	***11/10/11***

Project Description : ***Analysis of one aqueous sample for NPDES.***

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 114817. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

David R. Alkema

David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *114817*

Date Reported : *11/10/11*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: ***Analysis of one aqueous sample for NPDES.***

Sample ID : ***"Effluent"***

Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *11/01/11*
Sample Time : *0724*

Date Received : *11/02/11*
Sample Type : *aqueous*
KAR Sample No. : *114817-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Cyanide, available	*<0.002*	*mg/L*	*OIA-1677*	*11/08/11*	*JHB*	

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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : ***Ms. Amy Vail***

KAR Project No. :	***113221***
Date Reported :	***08/16/11***
Date Activated :	***08/02/11***
Date Due :	***08/16/11***
Date Validated :	***08/16/11***

Project Description : ***Analysis of one aqueous sample for NPDES.***

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 113221. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

David R. Alkema
David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : ***113221***

Date Reported : ***08/16/11***

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: ***Analysis of one aqueous sample for NPDES.***

Sample ID : ***"Effluent"***

Sampled By : *AJV of City of Cadillac, Water Resources*

Sample Date : *08/01/11*

Sample Time : *1115*

Date Received : *08/02/11*

Sample Type : *aqueous*

KAR Sample No. : ***113221-01***

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Cyanide, available	*<0.002*	*mg/L*	*OIA-1677*	*08/15/11*	*JHB*	

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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : ***Ms. Amy Vail***

KAR Project No. :	***111717***
Date Reported :	***05/12/11***
Date Activated :	***05/03/11***
Date Due :	***05/17/11***
Date Validated :	***05/12/11***

Project Description : ***Analysis of one aqueous sample for NPDES Permit.***

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 111717. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

David R. Alkema

David R. Alkema
Laboratory Manager

KAR Laboratories, Inc. maintains Full Certification status for Bacteriology, Inorganics, Regulated Organics and Synthetic Organics through USEPA, Michigan Department of Public Health and Indiana State Department of Health.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *111717*

Date Reported : *05/12/11*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: ***Analysis of one aqueous sample for NPDES Permit.***

Sample ID : ***"Effluent"***

Sampled By : *AJV of City of Cadillac, Water Resources*

Sample Date : *05/02/11*

Sample Time : *0901*

Date Received : *05/03/11*

Sample Type : *aqueous*

KAR Sample No. : *111717-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Cyanide, available	*<2*	*ug/L*	*OIA-1677*	*05/10/11*	*JHB*	

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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : ***Ms. Amy Vail***

KAR Project No. :	***110478***
Date Reported :	***02/08/11***
Date Activated :	***02/03/11***
Date Due :	***02/17/11***
Date Validated :	***02/07/11***

Project Description : ***Analysis of one aqueous sample for NPDES Permit.***

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 110478. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

David R. Alkema

David R. Alkema
Laboratory Manager

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : *110478*

Date Reported : *02/08/11*

Attest: David R. Alkema
David R. Alkema, Lab Manager

Project Description: *Analysis of one aqueous sample for NPDES Permit.*

Sample ID : ***"Effluent"***

Sampled By : *AV of City of Cadillac, Water Resources*
Sample Date : *02/01/11*
Sample Time : *1012*

Date Received : *02/03/11*
Sample Type : *aqueous*
KAR Sample No. : *110478-01*

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Cyanide, available	<2	ug/L	OIA-1677	02/03/11	JHB	